Julian Ockelmann

Contact information

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Personal Information

Address
Ueberlandstrasse 375,
8051 Zürich, Switzerland

Birthday, Place
18.05.1998, Frankfurt am Main

Nationality
German

Language Skills

German native proficiency

English native proficiency

French basic proficiency

Japanese basic proficiency

Technical Skills

R expert

PsychoPy advanced

BrainVisionAnalyzer advanced

Python proficient

Matlab proficient

Praat proficient

JavaScript intermediate

References

Prof. Dr. Nathalie Giroud

Prof. Dr. Martin Meyer

Experience

Research Doctorate (100%), University of Zurich

February 2023 – July 2026

- Conceptualization and Implementation of Neuroscientific Experiments
- Development of health-related Mobile Applications
- Development of an Open-Access Cognitive Screening Battery
- Longitudinal Modelling, Structural Equation Modelling, Linear Mixed Effect Modelling, ML-based Cluster Inference
- Analysis of Cortical and Subcortical speech-related Brain Signals

Teaching Assistant (40%), University of Zurich

February 2023 - July 2023

The Neural Foundation of Speech

- Teaching & Preparing Lectures
- Course Administration
- Grading final Examination

Research Assistant (20-50%), University of Zurich

October 2021 - February 2023

Computational Neuroscience of Speech & Hearing (50%)
July 2022 - December 2022

- Development of mHealth Application
- Longitudinal Data analysis in R
- In-Depth Literature Review

Plasticity and Learning (25%)
September 2020 - September 2021

- Advanced Data Analysis (R/Matlab)
- Research Database Maintenance

Methods of Plasticity Research (40%)
September 2021 - May 2022

- EEG Measurements
- ‘Big Data’ Analysis in R
- Application Construction in R
- Recruitment

Clinical Assistant (100%), Various Institutions
January 2019 - October 2019

Hospital zum Heiligen Geist (100%) - Clinic of Psychosomatic Medicine

Kinderzentrum KidZ (100%) - Child and Adolescent Psychiatry

Education

PhD - Cognitive Neuroscience, University of Zurich
February 2023 – July 2026

- **Dissertation:** *Benefits of auditory-cognitive training for hearing rehabilitation and cognitive ability in hearing-impaired older adults.*

PhD Fellow - Max Planck LIFE Program, University of Zurich
October 2023 – July 2026

- **Extracurricular:** PhD Fellow Speaker, LIFE Zurich

Research Stay – Department of Psychology, University of Michigan
April 2025 - May 2025

- **Project:** Identification of parietal alpha modulation as a function of cognitive training, using EEG data of 54 older adults with hearing loss.

Master of Science - Psychology, University of Zurich
September 2020 - February 2023

- **Thesis:** *The role of working memory capacity for tinnitus and speech-in-noise compehension in older adults with untreated hearing loss.*
- **Major 120 ECTS:** Psychology
- **Additional 36 ECTS:** Computer Science, Psychology
- **Focus:** Clinical Neuropsychology & Health Psychology
- **Extracurricular:** Member of Language & Medicine Competence Center

Supplementary Studies - Computational Science, Goethe University Frankfurt
March 2020 - August 2020

- **Focus:** Data Science & Database Management

Bachelor of Science - Psychology, HS Fresenius Frankfurt
March 2017 - January 2020

- **Thesis:** *Influence of physiological arousal state on intelligence test performance, considering self-efficacy expectations and anxiety levels.*
- **Major 180 ECTS**: Psychology
- **Focus:** Health Psychology & Clinical Psychology
- **Extracurricular:** Student Representative, Voluntary Tutor (Biopsychology)

Abitur Certification, Georg-Büchner-Gymnasium Bad Vilbel
September 2008 - June 2016

- **Advanced Classes:** English, History, Chemistry
- **Extracurricular:** Student Representative, Student Press Member

Publications

Published/Submitted

- **Ockelmann, J. F**., Stropahl, M., Scherpiet, S. & Giroud, N. R. (2025)
  Personalized and gamified auditory-cognitive training improves naturalistic speech-in-noise comprehension in older adults with hearing loss. *npj Science of Learning*, *10*(1), 80. https://doi.org/10.1038/s41539-025-00369-4
- Niharika, M. K., **Ockelmann, J. F.**, Meyer, M., & Giroud, N. R. (2025). *Identifying predictors of speech-in-noise perception in older adults with hearing impairment: The key role of working memory. BMC Geriatrics* (2025), https://doi.org/10.1186/s12877-025-06841-5
- Elmer, S., Frei V., **Ockelmann, J.F.**, Giroud, N.R. (2025). *Breaking Down the Ear-Brain Dichotomy: The Effects of Age-Related Hearing Loss on the Cortical Language System.* NeuroImage (2025), https://doi.org/10.1016/j.neuroimage.2025.121563
- **Ockelmann, J.F.**, Elmer, S., & Giroud, N. (2026). Cognition governs whether neural alpha indexes speech-in-noise outcomes. *(Nature Communications Psychology, Under Revision, 2nd Round)* https://doi.org/10.21203/rs.3.rs-8552497/v1
- **Ockelmann, J.F.**, Elmer, S., Reuter-Lorenz, P., & Giroud, N. (2026). *Auditory-Cognitive Training Is Associated with Selective Reorganization of Alpha-Band Connectivity During Speech-in-Noise Comprehension. (NeuroImage, Under Revision, 1st Round)* https://dx.doi.org/10.2139/ssrn.6639340
- **Ockelmann, J.F.***, Straus, J.*, Bolt, E., Elmer, S., & Giroud, N. (2026). Cognition governs whether neural alpha indexes speech-in-noise outcomes. *(Brain & Language, Under Revision, 2nd Round)* http://dx.doi.org/10.2139/ssrn.6560619

In Preparation

- **Ockelmann, J. F.**, Ulken, J. L., Baur, S., & Giroud, N. R.
  WOMBAT - A multilingual open-access Working Memory Test Battery for python. [in Preparation]
- **Ockelmann, J.F.***, Wildgruber, A.*, Elmer, S., Giroud, N. R.
  Auditory Assessment of Hidden Hearing Loss through Auditory Brainstem Responses and ERP Extraction in Older Adults. [in Preparation]

Ongoing

- Kretzschmar, L., **Ockelmann, J. F.,** Andratschke, N.
  AI-based autosegmentation of vestibular schwannoma in voxel space. [in Preparation]
- **Ockelmann, J. F.*,** Kretzschmar, L.*, Andratschke, N.
  Tumour growth during observation predicts hearing decline before stereotactic radiotherapy for vestibular schwannoma. [in Preparation]

Teaching & Initiative

Teaching

- Lecture – The neural foundation of speech, Department of Computational Linguistics; Guest Lecturer
  February 2024 - May 2024
- Lecture – The neural foundation of speech, Department of Computational Linguistics; Guest Lecturer, Exam Conceptualization & Evaluation
  February 2023 - May 2023

Initiative

- Co-organization of 3-day Fall LIFE Academy in Zurich (82 international faculty and fellows), University of Zurich, Switzerland.
  October 2025

- Organization of a workshop for designing cognitive neuroscience experiments in PsychoPy, University of Zurich, Switzerland.
  May 2024
- LIFE Fellow Speaker Zurich, International Max Planck Research School on the Life Course
  November 2023 - Present
- Co-organization of 3-day Fall LIFE Academy in Zurich (70 international faculty and fellows), University of Zurich, Switzerland.
  November 2023

Supervision

MSc Theses

- *Auditory Assessment of Hidden Hearing Loss through Auditory Brainstem Responses and ERP Extraction in Older Adults.*
- *Mu Oscillations and Inhibitory Control in ADHD During a Conversational Turn-Taking Paradigm.*
- *Temporal Microstate Dynamics in Speech-in-Noise Comprehension: Effects of Auditory-Cognitive Training and Hearing Aids in Older Adults with Hearing Loss.*
- *The Role of Subjective and Objective Measurements in Assessing Hearing Loss: An Examination of the Reliability of Speech, Spatial, and Qualities of Hearing (SSQ) and the Influence of Cognitive Factors.*
- *Variability of Neural Alpha Power during Speech Perception in Older Adults with Hearing Impairment.*

Programming Projects

- *Java Script integration of WoMBaT Subtests for Online Use.*

Societies & Memberships

- Member; Language & Medicine Competence Centre, Switzerland
- Member; Swiss Reproducibility Network (SwissRN) Academy, Switzerland
- Fellow; Max Planck Society IMPRS LIFE Program, Germany, Switzerland, United States

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Selected Contributions & Talks

- **Ockelmann, J. F.**, Rosemann, S. (2025) Symposium Chairing/Symposium Talk – *Cognition, Hearing & Tinnitus: Mechanisms and Implications.* Presented at PUG Conference, June 20th, 2025, Würzburg, Germany
- **Ockelmann, J.F.**, Giroud, N.R. (2025) Magazine Article – *Hearing meets Cognition: Modern Hearing Interventions up close.* Fachzeitschrift Hörakustik, July, 2025, Heidelberg, Germany.
- **Ockelmann, J. F.** (2025) Conference Talk - *Auditory-cognitive training improves naturalistic speech-in-noise comprehension in older adults.* Presented at Max Planck LIFE Michigan Academy, May 14th, 2025, Ann Arbor, United States.
- **Ockelmann, J. F.** (2023) Conference Talk – *WOMBAT: An open-access research tool for working memory assessment.* Presented at Language & Medicine Competence Centre, March 23th, 2024, Zurich, Switzerland.
- **Ockelmann, J. F.** (2023) Conference Talk – *Improving speech-in-noise understanding by means of auditory cognitive training.* Presented at Language & Medicine Competence Centre, April 29th, 2023, Zurich, Switzerland.